

Text-to-Schema with Large Language Models and Bayesian Networks (The supplementary materials)

1st Hoa Le Thi

Lyon 1 University & CNRS LIRIS

Lyon, France

thi-hoa.le@etu.univ-lyon1.fr

2nd Angela Bonifati

Lyon 1 University & CNRS LIRIS & IUF

Lyon, France

angela.bonifati@univ-lyon1.fr

3rd Andrea Mauri

Lyon 1 University & CNRS LIRIS

Lyon, France

andrea.mauri@univ-lyon1.fr

ER schema quality metrics for various configurations, LLMs, and prompts. Best overall configuration in grey

Config	LLM	Prompt	Basic Counts	Complexity	Redundancy (%) ↓	Well-Formedness (%) ↑
			E R A Total	R/E N:M (%) A/E	Dup Similar	Ent w/ A Valid Card Not Isolated
-0.5+1.0	GPT	FewShot	5.79 5.29 24.12 35.19	0.88 6.84 4.25	0.0 0.5	100.0 97.59 100.0
	LLAMA-3	FewShot	3.34 2.29 1.23 6.85	0.68 8.29 0.31	0.0 0.51	100.0 100.0 100.0
	GPT	ZeroShot	4.97 4.21 20.51 29.7	0.71 6.75 3.57	0.0 0.51	100.0 78.0 100.0
	LLAMA-3	ZeroShot	5.28 5.03 5.96 16.28	0.92 25.06 1.05	0.0 0.48	100.0 100.0 100.0
	GPT	ToT	8.44 8.86 46.32 63.62	1.04 3.48 5.51	0.0 0.29	100.0 99.12 100.0
	LLAMA-3	ToT	3.43 2.3 14.69 20.42	0.67 4.87 4.32	0.0 0.24	100.0 100.0 100.0
0.5+1.0	GPT	FewShot	5.73 5.3 24.27 35.3	0.88 6.86 4.28	0.0 0.5	100.0 97.19 100.0
	LLAMA-3	FewShot	4.85 4.37 21.88 31.1	0.87 6.67 4.61	0.0 0.23	100.0 100.0 100.0
	GPT	ZeroShot	4.96 3.93 20.62 29.51	0.68 3.61 3.7	0.0 0.32	100.0 75.00 100.0
	LLAMA-3	ZeroShot	6.34 6.08 24.79 37.22	0.94 18.03 4.02	0.0 0.51	100.0 100.0 100.0
	GPT	ToT	8.44 8.86 46.31 63.6	1.04 3.48 5.51	0.0 0.29	100.0 99.12 100.0
	LLAMA-3	ToT	6.53 6.37 25.89 38.79	0.97 10.99 4.03	0.0 0.32	100.0 100.0 100.0
1.0+3.0	GPT	FewShot	5.54 5.18 23.93 34.65	0.87 5.87 4.29	0.0 0.52	100.0 97.0 100.0
	LLAMA-3	FewShot	5.14 4.91 20.95 31.01	0.88 6.30 3.88	0.0 0.72	100.0 94.25 100.0
	GPT	ZeroShot	4.22 3.93 17.55 25.7	0.68 3.61 3.2	0.0 0.27	100.0 75.00 100.0
	LLAMA-3	ZeroShot	6.34 6.11 24.84 37.28	0.95 17.84 4.03	0.0 0.51	100.0 100.0 100.0
	GPT	ToT	8.29 6.78 46.08 61.15	0.8 1.78 5.56	0.0 0.26	100.0 98.77 100.0
	LLAMA-3	ToT	6.14 3.89 25.14 35.16	0.63 2.37 4.15	0.0 0.26	100.0 100.0 100.0